

The missing factor is logarithmic A sharp solution of the strict small-smoothness Morrey entropy gap

Autonomous solve-first investigation, lane 12

13 August 2026

Abstract

Haroske and Skrzypczak left an epsilon gap for entropy numbers of compact embeddings between proper smoothness Morrey spaces. In the strict small-smoothness range we close it: the exact order is

$$k^{-\alpha}(\log(e+k))^\alpha,$$

not the previously suggested pure power. The lower bound comes from a large dyadic code with one spike in each mesoscopic block. The matching upper bound follows by retaining the finite-level Schuett estimate already present in the source proof and distributing the entropy budget directly across wavelet levels. At the limiting face, the same packing gives the new lower bound $k^{-D}(\log(e+k))^D$, but the matching endpoint upper bound, which may depend on the fine indices, is not claimed here.

1 Source question and status

The source is D. D. Haroske and L. Skrzypczak, *Entropy numbers of compact embeddings of smoothness Morrey spaces on bounded domains*, arXiv:1902.04945, later J. Approx. Theory 256 (2020), 105424 [1]. Its Theorems 3.13(ii) and 4.4(ii) prove

$$ck^{-\alpha} \leq e_k(T) \leq c_\varepsilon k^{-\alpha+\varepsilon} \quad (\varepsilon > 0), \quad (1)$$

and Remark 4.12 asks whether a direct covering argument can close the gap. The exact printed passage is shown in Figure 1.

The result below fully determines the entropy order in the open strict range. Because the limiting equality case remains only partially resolved, this packet is conservatively classified as a substantial partial solution to the whole of Remark 4.12.

2 Notation

Fix $0 < p_i \leq u_i < \infty$ and $0 < q_i \leq \infty$, $i = 1, 2$, and put

$$D = \frac{1}{p_1} - \frac{1}{p_2}, \quad a = 1 - \frac{p_1}{u_1}, \quad \Delta = \frac{s_1 - s_2}{d}, \quad \beta = \Delta - \frac{p_1}{u_1}D, \quad \text{quad} \alpha = \frac{\beta}{a}. \quad (2)$$

In the nonclassical situation of the source,

$$\frac{p_1}{u_1}D > \max \left\{ 0, \frac{1}{u_1} - \frac{1}{u_2} \right\}, \quad (3)$$

so $p_1 < p_2$, $p_1 < u_1$, and $p_2/u_2 > p_1/u_1$.

For completeness, at a fixed dyadic level j let $n = 2^{jd}$ and let $m_{u,p}^{(j)}$ be $\mathbb{R}^n$ with norm

$$\|x\|_{m_{u,p}^{(j)}} = \max_{0 \leq \nu \leq j} \max_{Q_{\nu,r} \subset Q} N_\nu^{1/u-1/p} \left(\sum_{Q_{j,m} \subset Q_{\nu,r}} |x_m|^p \right)^{1/p}, \quad N_\nu = 2^{d(j-\nu)}. \quad (4)$$

Corollary 4.9. Let Ω be a bounded domain in $\mathbb{R}^d$, $s_i \in \mathbb{R}$, $0 < q_i \leq \infty$, $0 < p_i \leq u_i < \infty$, $i = 1, 2$, and [4.7](#) satisfied. Assume that $p_2 = u_2$.

(i) If $\frac{s_1 - s_2}{d} > \left(\frac{1}{p_1} - \frac{1}{p_2}\right)_+$, then

$$e_k(\text{id} : \mathcal{A}_{u_1, p_1, q_1}^{s_1}(\Omega) \hookrightarrow A_{p_2, q_2}^{s_2}(\Omega)) \sim k^{-\frac{s_1 - s_2}{d}}.$$

(ii) If $\frac{p_1}{u_1} \left(\frac{1}{p_1} - \frac{1}{p_2}\right) < \frac{s_1 - s_2}{d} \leq \frac{1}{p_1} - \frac{1}{p_2}$, then there exists some $c > 0$ and for any $\varepsilon > 0$ some $c_\varepsilon > 0$ such that for all $k \in \mathbb{N}$,

$$ck^{-\frac{u_1}{u_1 - p_1} \left(\frac{s_1 - s_2}{d} - \frac{p_1}{u_1} \left(\frac{1}{p_1} - \frac{1}{p_2}\right)\right)} \leq e_k(\text{id}) \leq c_\varepsilon k^{-\frac{u_1}{u_1 - p_1} \left(\frac{s_1 - s_2}{d} - \frac{p_1}{u_1} \left(\frac{1}{p_1} - \frac{1}{p_2}\right)\right) + \varepsilon}.$$

Proof. This follows from Theorems [4.2](#) and [4.4](#) with $p_2 = u_2$. $\square$

Corollary 4.10. Let Ω be a bounded domain in $\mathbb{R}^d$, $1 \leq r \leq \infty$, $0 < p \leq u < \infty$, and $s > d \frac{p}{u} \left(\frac{1}{p} - \frac{1}{r}\right)_+$.

(i) If $p \geq r$, then

$$e_k(\text{id} : \mathcal{A}_{u, p, q}^s(\Omega) \hookrightarrow L_r(\Omega)) \sim k^{-\frac{s}{d}}. \quad (4.8)$$

(ii) If $p < r$ and $d \frac{p}{u} \left(\frac{1}{p} - \frac{1}{r}\right) < s \leq d \left(\frac{1}{p} - \frac{1}{r}\right)$, then there exists some $c > 0$ and for any $\varepsilon > 0$ some $c_\varepsilon > 0$ such that for all $k \in \mathbb{N}$,

$$ck^{-\frac{u}{u - p} \left(\frac{s}{d} - \frac{p}{u} \left(\frac{1}{p} - \frac{1}{r}\right)\right)} \leq e_k(\text{id} : \mathcal{A}_{u, p, q}^s(\Omega) \hookrightarrow L_r(\Omega)) \leq c_\varepsilon k^{-\frac{u}{u - p} \left(\frac{s}{d} - \frac{p}{u} \left(\frac{1}{p} - \frac{1}{r}\right)\right) + \varepsilon}, \quad (4.9)$$

where $L_\infty(\Omega)$ can be replaced in all cases by $C(\Omega)$.

Proof. The result follows from Corollary [4.9](#) together with the well-known embeddings $B_{r,1}^0(\Omega) \hookrightarrow L_r(\Omega) \hookrightarrow B_{r,\infty}^0(\Omega)$ and the multiplicativity of entropy numbers [2.11](#). $\square$

Remark 4.11. We obtained in [16](#), [50](#) some first results for approximation numbers of the embedding id_A . In this context we refer to [1](#) Sect. 6] where also the periodic case and more general Morrey type spaces are studied.

Remark 4.12. Thanks to one reviewer we were pointed to the recent preprint [25](#) which is based on [5](#). It might well be that using this approach one could circumvent our above interpolation argument and thus seal the ε -gap in the exponents in Theorems [3.13](#)(ii) and [4.4](#)(ii), respectively. We leave it as an open question here.

Acknowledgment

Figure 1: The epsilon-gap estimate and open question on printed p. 18 of arXiv:1902.04945v2.

The associated bounded-cube Besov–Morrey sequence norm is

$$\|x\|_{\tilde{n}_{u,p,q}^\sigma} = \left\| \left(2^{j(\sigma - d/u)} \|x^{(j)}\|_{m_{u,p}^{(j)}} \right)_{j \geq 0} \right\|_{\ell_q}. \quad (5)$$

As in the source, $\sigma_i = s_i + d/2$, so only their difference matters.

3 Proof intuition

The published lower bound treats level j as if it had about n^a effective coordinates. That count is right, but each effective coordinate can be placed in about n^{1-a} different fine cubes. Coding these choices produces $\exp(cn^a \log n)$, rather than merely $\exp(cn^a)$, separated vectors. This forces the logarithm in the answer.

For the upper bound one should not replace the finite-level entropy estimate by an $L_{r,\infty}^{(e)}$ ideal norm, because taking $1/r > D$ is exactly where the epsilon is introduced. The source proves a sharper Schuett estimate immediately before that step. Its transition also occurs at $n^a \log n$. In the strict range $\beta < aD$, entropy budgets assigned to successive lower levels form a geometric series. This gives a cover of radius $N^{-\beta}$ using $O(N^a \log N)$ bits, precisely matching the code.

4 The strict-range theorem

Theorem 1 (sharp logarithmic order). *Let $\Omega \subset \mathbb{R}^d$ be any bounded domain, let $\mathcal{A} \in \{\mathcal{N}, \mathcal{E}\}$, and assume (3) and*

$$\frac{p_1}{u_1}D < \Delta < D. \quad (6)$$

For the compact embedding

$$T : \mathcal{A}_{u_1, p_1, q_1}^{s_1}(\Omega) \longrightarrow \mathcal{A}_{u_2, p_2, q_2}^{s_2}(\Omega)$$

one has, for every $k \in \mathbb{N}$,

$$e_k(T) \asymp k^{-\alpha}(\log(e+k))^\alpha, \quad \alpha = \frac{u_1}{u_1 - p_1} \left(\Delta - \frac{p_1}{u_1}D \right). \quad (7)$$

The constants may depend on the fixed parameters and on Ω , but not on k .

We prove the sequence-space statement first.

Lemma 2 (finite-level upper estimate). *Set*

$$\gamma = \frac{1}{u_2} - \frac{p_1}{u_1 p_2} > 0, \quad S_j : m_{u_1, p_1}^{(j)} \longrightarrow m_{u_2, p_2}^{(j)}.$$

Uniformly in $n = 2^{jd}$ and $m \geq 1$,

$$e_m(S_j) \lesssim n^\gamma \min \left\{ 1, \left(\frac{n^a \log(e + n/m)}{m} \right)^D \right\}, \quad 1 \leq m \lesssim n, \quad (8)$$

$$e_m(S_j) \lesssim n^{1/u_2 - 1/u_1} 2^{-cm/n}, \quad m \gtrsim n. \quad (9)$$

Consequently the level operator induced by $\tilde{n}_{u_1, p_1, q_1}^{\sigma_1} \rightarrow \tilde{n}_{u_2, p_2, q_2}^{\sigma_2}$ satisfies

$$e_m(T_j) \lesssim n^{-\beta} \min \left\{ 1, \left(\frac{n^a \log(e + n/m)}{m} \right)^D \right\}, \quad m \lesssim n, \quad (10)$$

$$e_m(T_j) \lesssim n^{-\Delta} 2^{-cm/n}, \quad m \gtrsim n. \quad (11)$$

Proof. Equation (3.35) in Step 1 of Proposition 3.12 of the source gives, for the renormalized target block m_{u_2, p_2}^* ,

$$e_{2m-1}(m_{u_1, p_1}^{(j)} \rightarrow m_{u_2, p_2}^*) \lesssim n^{aD} \left(\frac{\log(1 + n/m)}{m} \right)^D$$

in the Schuett range. Equation (3.32) says that returning to the original target multiplies entropy numbers by n^γ . Taking the minimum with $\|S_j\| \lesssim n^\gamma$ yields (8); the last branch of (3.35), equivalently the standard finite-dimensional volumetric estimate, gives (9).

The ratio of the level weights in (5) is $n^{-(\Delta - 1/u_1 + 1/u_2)}$. Since

$$-\Delta + \frac{1}{u_1} - \frac{1}{u_2} + \gamma = -\beta, \quad -\Delta + \frac{1}{u_1} - \frac{1}{u_2} + \frac{1}{u_2} - \frac{1}{u_1} = -\Delta,$$

the weighted estimates follow. $\square$

Lemma 3 (multilevel upper bound). *Under (6), for $N = 2^{Jd}$,*

$$e_{CN^a(1+J)}(\tilde{n}_{u_1, p_1, q_1}^{\sigma_1} \circ \tilde{n}_{u_2, p_2, q_2}^{\sigma_2}) \lesssim N^{-\beta}. \quad (12)$$

Proof. Choose

$$0 < \eta < D - \Delta = aD - \beta \quad (13)$$

and put $R = N^{-\beta}$. The tail $j > J$ is covered by the zero center: from (10) with $m = 1$ and the fact that every normalized source level is at most one,

$$\left\| (T_j x^{(j)})_{j>J} \right\|_{\ell_{q_2}} \lesssim \left\| (2^{-jd\beta})_{j>J} \right\|_{\ell_{q_2}} \lesssim R,$$

with the evident supremum interpretation when $q_2 = \infty$.

For $j \leq J$, write $r = J - j$, $n_j = N2^{-dr}$, and prescribe the level error

$$R_j = R2^{-\eta dr}. \quad (14)$$

Let

$$c_0 = a - \frac{\beta + \eta}{D} > 0.$$

Whenever the following number is at most a fixed multiple of n_j , choose

$$m_j = \left\lceil CN^a(1+j)2^{-c_0 dr} \right\rceil. \quad (15)$$

Since $\log(e + n_j/m_j) \lesssim 1 + j$, the bracket in (10) is at most a constant times $2^{-(\beta+\eta)dr/D}$. Hence $e_{m_j}(T_j) \lesssim R_j$. Moreover,

$$\sum m_j \lesssim N^a \sum_{r \geq 0} (1 + J - r) 2^{-c_0 dr} \lesssim N^a(1 + J). \quad (16)$$

It remains to audit the lower levels where (15) exceeds n_j . There use (11) and take

$$m_j = \left\lceil Cn_j \left(1 + \log_+ \frac{n_j^{-\Delta}}{R_j} \right) \right\rceil. \quad (17)$$

This again makes $e_{m_j}(T_j) \lesssim R_j$. The condition that (15) exceeds n_j implies

$$n_j \lesssim N^\kappa (1 + J)^{1/(1-c_0)}, \quad \kappa = \frac{a - c_0}{1 - c_0} < a.$$

The n_j are geometric, while the logarithm in (17) is $O(1 + J)$. Thus the sum of all budgets in this lower-level part is

$$O(N^\kappa (1 + J)^{2+1/(1-c_0)}) = o(N^a(1 + J)).$$

Take the Cartesian product of the chosen level covers. It uses at most $2^{CN^a(1+J)}$ centers. By (14), its target radius is

$$\|(R_j)_{j \leq J}\|_{\ell_{q_2}} \lesssim R;$$

adding the tail changes only the constant. This proves (12). $\square$

Lemma 4 (dyadic code lower bound). *Under the nonclassical hypotheses and merely $\beta > 0$,*

$$e_{cn^a \log n}(\tilde{n}_{u_1, p_1, q_1}^{\sigma_1} o \tilde{n}_{u_2, p_2, q_2}^{\sigma_2}) \gtrsim n^{-\beta}, \quad n = 2^{jd}, \quad (18)$$

for all sufficiently large j .

Proof. Put $\nu = \lfloor aj \rfloor$, $s = 2^{d\nu}$, and $L = 2^{d(j-\nu)}$, so $s \asymp n^a$, $L \asymp n^{1-a}$, and $n = sL$. Partition the fine cubes at level j into the s dyadic cubes at level ν . For each word $\omega \in \{1, \dots, L\}^s$, let x^ω be the indicator that chooses the ω_t -th fine cube in parent t .

Every x^ω lies in the unit ball of $m_{u_1, p_1}^{(j)}$. Indeed, a dyadic block containing M fine cubes contains at most M^a chosen cubes. For a block below the parent scale this count is at most one. For a block above it the assertion is equivalent to $2^{d(\nu-\mu)} \leq 2^{ad(j-\mu)}$, which follows from $\nu \leq aj$. Substitution in (4) proves the claim.

A greedy code in the alphabet of size L gives a set $\mathcal{C} \subset \{1, \dots, L\}^s$ with Hamming distance at least $s/2$ and

$$|\mathcal{C}| \geq 2^{cs \log_2 L}. \quad (19)$$

For reference, a Hamming ball of radius $s/2$ has at most $2^s L^{s/2}$ words, so the elementary greedy deletion argument proves (19).

If $\omega, \omega' \in \mathcal{C}$ are distinct, their indicators differ in at least s fine coordinates. Testing the target norm on the whole cube therefore gives

$$\|x^\omega - x^{\omega'}\|_{m_{u_2, p_2}^{(j)}} \gtrsim n^{1/u_2 - 1/p_2} s^{1/p_2} \text{asympt} n^\gamma. \quad (20)$$

Multiply every vector by the source normalization $n^{-(\sigma_1/d - 1/u_1)}$. They now lie in the unit ball of the full source sequence space and are supported on one level. After the target weight is applied, (20) becomes

$$n^{-\Delta + 1/u_1 - 1/u_2} n^\gamma = n^{-\beta}.$$

The packing cardinality is $2^{cn^a \log n}$, proving (18) by the definition of entropy numbers. $\square$

Proof of Theorem 1. The quantities $K_J = 2^{Jda}(1 + J)$ have bounded successive ratios. Inverting (12) and (18), and using monotonicity of entropy numbers between adjacent K_J , yields

$$e_k(\tilde{n}_{u_1, p_1, q_1}^{\sigma_1} \tilde{o} n_{u_2, p_2, q_2}^{\sigma_2}) \asymp k^{-\beta/a} (\log(e + k))^{\beta/a}.$$

Indeed, $K_J \asymp n^a \log n$ and $n^{-\beta} \asymp K_J^{-\beta/a} (\log K_J)^{\beta/a}$.

The two wavelet-factorization diagrams in Section 4 of the source transfer both entropy inequalities between bounded-cube sequence spaces and Besov–Morrey spaces on every bounded domain. Finally, the elementary embeddings

$$\mathcal{N}_{u, p, \min(p, q)}^s \hookrightarrow \mathcal{E}_{u, p, q}^s \hookrightarrow \mathcal{N}_{u, p, \infty}^s$$

and multiplicativity of entropy numbers give the same two-sided order for the Triebel–Lizorkin–Morrey scale. The formula is independent of the fine indices, so the two sandwiches agree. $\square$

5 Endpoint consequence and limitation

Proposition 5. *Under (3), if $\Delta = D$, then for both function-space scales and every bounded domain,*

$$e_k(T) \gtrsim k^{-D} (\log(e + k))^D. \quad (21)$$

In particular, $e_k(T) \asymp k^{-D}$ is impossible.

Proof. Here $\beta = aD$. Lemma 4 and the same adjacent-level inversion give (21); the wavelet transfer is unchanged. $\square$

At this equality face, $\beta = aD$, so the geometric factor in the budget (15) vanishes. Many levels can contribute at the same order, and a sharp cover may depend on q_1, q_2 , just as anticipated in Remark 4.7 of the source. The packet therefore makes no matching endpoint upper-bound claim.

6 Verification and novelty audit

The following points were checked independently during the solve-first pass.

- The exponent identities in Lemma 2 were expanded symbolically by hand in both directions; the transition at $m = n$ agrees with the volumetric branch because $\beta + (1 - a)D = \Delta$.
- The source-ball audit in Lemma 4 checks every dyadic scale, not only the whole cube. The target separation uses a legitimate whole-cube term in the supremum.
- The upper budget was split exactly where the Schuett-range allocation exceeds the actual dimension. The strict inequality $\Delta < D$ is used precisely to obtain $c_0 > 0$ and $\kappa < a$.
- No numerical experiment is used as proof.

- On 13 August 2026, searches of the run indexes and bounded web searches for the exact title, the phrase “seal the epsilon-gap”, theorem numbers, Morrey sequence-space entropy, the authors, arXiv:1904.04619, and later Morrey-embedding work found no prior closure of this question. The search did find the source journal version and later nuclearity/survey papers, none of which supplies this logarithmic theorem.

Human-review recommendation. Review as a candidate substantial partial theorem with a full proof for the strict range. The most useful verifier focus is the multilevel product-cover budget in Lemma 3; the finite-level inequality it uses is the source’s printed equation (3.35), and the lower code is elementary.

References

- [1] D. D. Haroske and L. Skrzypczak, *Entropy numbers of compact embeddings of smoothness Morrey spaces on bounded domains*, J. Approx. Theory 256 (2020), 105424; arXiv:1902.04945.